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Decoding Spotify: A Data Science Approach to Music

Research Questions:

1. Which audio features are most strongly associated with song popularity?
 - Result: None of the selected audio features show a meaningful relationship with popularity scores (standardized within genres using genre-level means and standard deviations) at the 95% confidence level; effect sizes are near zero, and the associated OLS models explain minimal variance.
2. Can song popularity be predicted using audio features and tracked metadata, and if so, which ones, or which combination of features provide the best prediction?
 - Result: Regression models using all audio features combined with metadata as predictors showed limited ability to predict song popularity or explain variance in data with the given predictors (Low R^2).
3. Do different genres exhibit distinct patterns in audio features?
 - Result: The heatmap shows audio features such as instrumentalness, valence, acousticness, and danceability vary systematically across genres, suggesting that different genres are characterized by distinct audio feature profiles.
4. How do explicit and non-explicit songs differ in popularity and audio characteristics?
 - Result: Explicit and non-explicit songs show statistically significant differences across several audio features, including popularity, energy, loudness, speechiness, instrumentalness, liveness, tempo, and time signature, indicating measurable separation between the two groups.

Motivation

Music streaming platforms like Spotify are widely used for both entertainment and discovery, and recommendation features such as personalized playlists and “popular songs” suggestions strongly influence what users listen to. While using Spotify in daily life, often while working on coursework and listening through features like DJ-X recommendations, it becomes noticeable that certain songs are repeatedly surfaced as “popular” or “relevant” within my own listening context, even songs from artists I’d never heard of but came to enjoy because they matched my musical interests.

This naturally raises the question of what measurable characteristics make a song likely to be considered popular or widely recommended. Although Spotify’s internal recommendation and ranking algorithms are proprietary and not directly observable, the underlying audio features (such as danceability, energy, tempo, and acoustics) are publicly available and provide an opportunity to study similar patterns independently.

By analyzing these features, this project aims to better understand whether song popularity can be explained or predicted using objective musical attributes, and how different genres or audio profiles contribute to what is considered “popular.” More broadly, this helps illustrate how data-driven recommendation systems may reflect underlying structure in music consumption, and whether popularity is primarily driven by measurable audio characteristics or more complex, less interpretable factors.

Datasets

- <https://www.kaggle.com/datasets/sarametaIa/spotify-analysis-and-visualization>
 - o 1879 x 18 dataset
 - o Columns: artist, song, duration_ms, explicit, year, popularity, danceability, energy, key, loudness, mode, speechiness, acousticness, instrumentalness, liveness, valence, tempo, genre
- <https://www.kaggle.com/datasets/maharshipandya/-spotify-tracks-dataset?resource=download>
 - o 89741 x 21 dataset
 - o Columns: track_id, artists, album_name, track_name, popularity, duration_ms, explicit, danceability, energy, key, loudness, mode, speechiness, acousticness, instrumentalness, liveness, valence, tempo, time_signature, track_genre
- <https://www.kaggle.com/datasets/melissamonfared/spotify-tracks-attributes-and-popularity>
 - o 89741 x 21 dataset
 - o Columns: Index, track_id, artists, album_name, track_name, popularity, duration_ms, explicit, danceability, energy, key, loudness, mode, speechiness, acousticness, instrumentalness, liveness, valence, tempo, time_signature, track_genre

Method

- Data Preparation (applies to all research questions)
 - Load each Spotify dataset into pandas DataFrames using `pd.read_csv()`.
 - Standardize column names across datasets (e.g., ensure consistent naming for artist, song, popularity, genre, audio features, etc.).
 - Select relevant columns: artist, song, duration_ms, explicit, year, popularity, danceability, energy, key, loudness, mode, speechiness, acousticness, valence, tempo, genre.
 - Remove rows with missing values in key numeric fields to ensure valid statistical and ML analysis.
 - Merge multiple datasets using (artist, song) as a composite key after resolving duplicates and formatting inconsistencies (case normalization, whitespace removal).
 - Turn genre and artist columns into list to consider all unique combinations (some columns may have a comma delimited list)
 - Filter to data entries containing top 20 most-occurring genres and top 500 most-occurring artists to ensure plentiful variation in data without being overwhelming
 - Standardize popularity scores relative to genre-level popularity means and standard deviation
 - Computed with the following formula and stored in column “popularity_z”: $P_{\text{Standardized}} = \frac{P_{\text{Raw}} - \mu_{\text{Genre}}}{\sigma_{\text{Genre}}}$.
- Q1
 - Create a copy of the dataset for visualization purposes.
 - Aggregate numeric predictor values by artist, producing artist-level averages for simpler visualization and modeling.
 - Construct a matplotlib figure with a grid of subplots (e.g., 2×4 layout).
 - Each subplot shows a scatterplot of one predictor (x-axis) against popularity_z (y-axis), with an OLS regression line fitted to the data.
 - For each predictor, fit a simple linear regression model (`statsmodels smf.ols()` function) and record the coefficient, R^2 value, and p-value to evaluate relationship strength and significance.
 - Compare relationships across predictors using both statistical metrics and visual trends.
 - Set y-axis labels only on leftmost plots for readability and use predictor names as subplot titles.

- Q2:

- Carry over predictor variables from Q1, and make two training sets, one with just these audio features as predictors, and the other including artist name and genre
 - Split both filtered datasets into training/test sets using an 80/20 split (`train_test_split()` from `scikit-learn`).
 - Explode the artist and genre columns into dummy variables in the designated training set for running the model with those variables as predictors
 - Run `StandardScaler()` from `sklearn` to standardize training and test sets before training models
- Run Models with just audio predictors first
 - Lasso Regression: From `sklearn.linear_model`, import `LassoCV` to run the model with `cv = 5` for K-fold cross validation, 5000 iterations maximum, random seed (42) for reproducibility, and `alphas = np.logspace(-4, 1, 50)`
 - Random Forest Regression: From `sklearn.ensemble`, import `RandomForestRegressor` to run the model with 300 trees was trained, using a fixed random seed (42) for reproducibility and parallel computation across all available CPU cores.
 - Multilayer Perceptron Regression Model: From `sklearn.neural_network`, import `MLPRegressor` to run a 2-layer model (128 neurons in layer 1, 64 in layer 2) with a `relu`-activation function, and fixed random seed (42) for reproducibility.
 - For each model, report the R^2 value and MSE of the model when ran against the testing dataset
 - Report non-zero coefficient count for Lasso regression too
- Rerun each of the models trained on the datasets with audio, artist, and genre next and compare results for any potential improvements or shifts in results

- Q3:

- Copy dataset and filter to just numeric features and genre, then standardize scaling with `MinMaxScaler()` from `sklearn.preprocessing`, group by genre, and simplify to scale-feature means within genres
 - Scale features: "danceability", "energy", "loudness", "mode", "speechiness", "acousticness", "instrumentalness", "liveness", "valence",
- Plot a heatmap of the numeric feature means against genres
 - Use “rocket” color palette with white borders to increase contrast and visibility
 - `Vmin = 0`, `vmax = 1` to match standardized scaling of numeric features
- Observe color shifts for numeric feature means by column; corresponds to the shift by genre; identify highly variable and highly consistent numeric features

- Q4

- Select all numeric features for graphing: "popularity", "duration_ms", "danceability", "energy", "key", "loudness", "mode", "speechiness", "acousticness", "instrumentalness", "liveness", "valence", "tempo", "time_signature"
- Create a grid of subplots (3×5 layout) to visualize relationships between each numeric audio feature and explicit content.
 - For each feature, fit an Ordinary Least Squares (OLS) regression model of the form $\text{feature} \sim \text{explicit}$ to estimate the effect of explicit content on the feature.
 - Extract the coefficient for explicit content and its corresponding p-value to quantify effect size and statistical significance.
 - Generate bar plots showing mean feature values grouped by explicit vs non-explicit songs.
 - Overlay statistical annotations (difference in means and p-values) onto each subplot.
- Observe results and note any statistically significant differences between feature means between non-explicit and explicit songs

EDA/7-number summary:

Var Name	Count	Mean	STD	Min	25%	50%	75%	Max
duration_ms	9697	239267.73	146077.425	8586.0	160000.0	222000.0	288600.0	4447520.0
energy	9697	0.636	0.292201	0.0	2.01e-05	0.414	0.901	1.0
tempo	9697	120.43	31.8046	0.0	96.473	120.622	140.045	220.525
valence	9697	0.378	0.26521	0.0	0.15	0.338	0.57	0.993
speechiness	9697	0.10941	0.17846	0.0	0.0375	0.0517	0.093	0.963
danceability	9697	0.4950	0.20029	0.0	0.346	0.516	0.65	0.98
acousticness	9697	0.34747	0.36797	0.0	0.00231	0.185	0.717	0.996
loudness	9697	-10.023	6.29022	-42.631	-12.919	-8.143	-5.474	3.156
popularity	9697	24.802	14.56384	0.0	15.0	22.0	35.0	83.0
popularity_z	9697	-1.01852	0.99902	-3.96378	-0.5980	-0.2544	0.43596	7.4607

Exploratory data analysis revealed that the original combined dataset (over 84,000 songs) contained substantial inconsistencies, including missing values, duplicate entries, and variability across sources. This led to a series of preprocessing steps to improve data quality and analytical reliability that substantially reduced the size of the final cleaned dataset (9697).

Another key takeaway from the exploratory analysis was the inconsistent scaling across numeric variables, with features operating on vastly different ranges and distributions. This motivated explicit feature standardization in Q3 to ensure comparability across variables and improve the reliability of downstream modeling and analysis. The scaling issue was also what inspired the creation of the 'popularity_z' column for comparing features against popularity scores without having to continually split the data by genre. All summary statistics reported above are computed from the fully cleaned and filtered dataset, ensuring that the observed distributions reflect the final analysis-ready sample.

Results

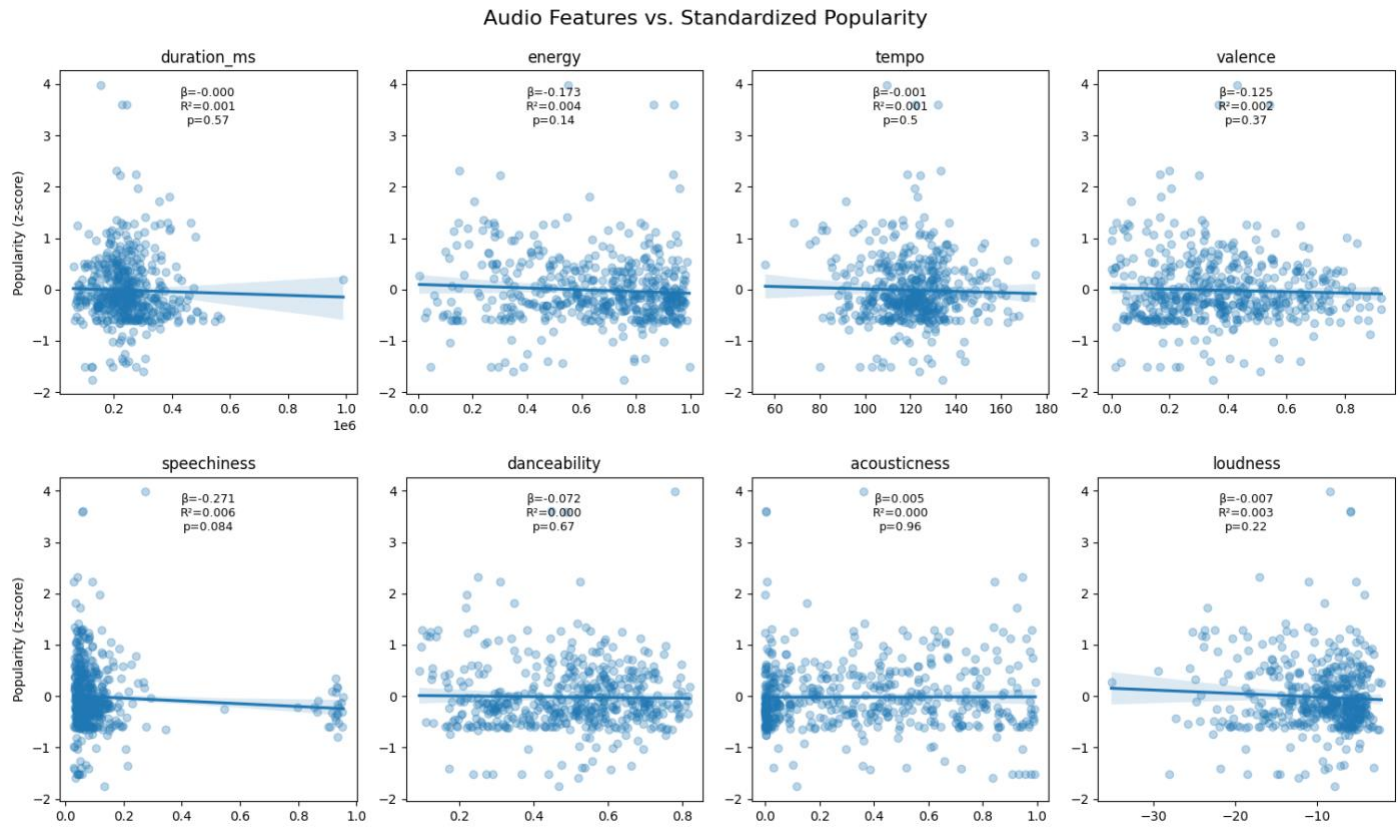


Figure 1

Q1: The results in Figure 1 suggest no statistically significant relationship between individual audio features and the standardized popularity score at the 95% confidence level (all p-values > 0.05). Additionally, the explanatory power of each single-feature model is extremely low; the highest observed R^2 value (speechiness) in the graph was approximately 0.006, suggesting no singular audio feature from this set can meaningfully account for variation in song popularity in this data. The results lead me to suspect popularity is likely driven by more complex structures, including interactions between multiple features and potentially nonlinear relationships that are not captured by simple univariate models. This is further supported by the consistently low R^2 values across all predictors, suggesting that explanatory power is distributed rather than concentrated in any one variable.

While it was not expected that any single feature would strongly dominate, the uniformly low R^2 values were still somewhat surprising, as a modest level of explanatory power was anticipated for at least some audio characteristics. Nonetheless, the results remain informative, as they motivate the use of multivariate and potentially nonlinear modeling approaches in subsequent analysis to better capture combined feature effects as a potential explanation for the relationship between audio features and song popularity; a relationship to be explored further in the next research question's experimentation.

```
===== AUDIO FEATURES ONLY =====  
  
=== LASSO ===  
R2: 0.0597  
MSE: 195.0998  
Nonzero coefficients: 8  
  
=== RANDOM FOREST ===  
R2: 0.2504  
MSE: 155.5441  
  
=== MLP ===  
R2: 0.1908  
MSE: 167.9074
```

Figure 2

```
===== AUDIO + ARTIST + GENRE =====  
  
=== LASSO ===  
R2: 0.1008  
MSE: 186.5873  
Nonzero coefficients: 13  
  
=== RANDOM FOREST ===  
R2: 0.3021  
MSE: 144.8065  
  
=== MLP ===  
R2: 0.1062  
MSE: 185.452
```

Figure 3

Q2: The Lasso regression model demonstrated limited predictive performance using the eight audio features according to Figure 2, achieving an R^2 of approximately 0.0597 and an MSE of 195.0998. This indicates that linear regularization and feature selection provided minimal improvement in explaining variation in song popularity. To further evaluate model performance, two nonlinear approaches were tested: a Random Forest Regressor and a Multilayer Perceptron (MLP). Both models produced only modest improvements over the linear baseline, with R^2 values remaining below approximately 0.30 and MSE values above 150. Notably, the MLP performed worse than the Random Forest, suggesting instability or limited learnable signal in the feature space.

The models were then re-evaluated using an expanded feature set that included artist identity and genre (supplemented into the model as dummy variables) in addition to the audio features. This resulted in only marginal performance gains as displayed in Figure 3, and in some cases no improvement at all, with the MLP again performing worse under the expanded feature set. This suggests that even with categorical context included, the predictive signal for popularity remains weak.

Overall, performance across all models remains consistently limited regardless of model complexity or feature expansion. This reinforces the earlier conclusion that popularity is not strongly determined by the available audio and metadata features alone but further clarifying that even a combined effect approach yields minimal results. This outcome suggests that additional external factors or more expressive feature engineering are likely required to meaningfully improve predictive performance, perhaps media factors or pop culture context that influences audience response beyond just technical data of the song itself.

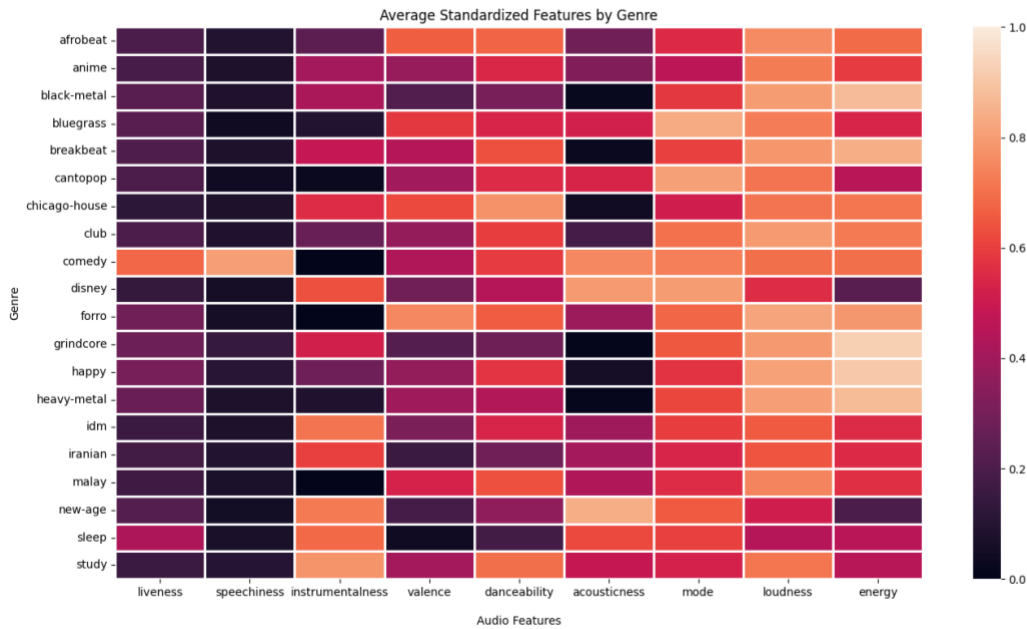


Figure 4

Q3: After graphing onto the heatmap in Figure 4, it appears the Spotify data does indeed demonstrate meaningful differences in audio features across genres, as initially suspected. The map illustrates the relationship between eight selected audio features and the 20 most frequently occurring genres in the dataset. For interpretability, all features were standardized to a common 0-1 scale prior to visualization, allowing relative differences to be compared directly across variables. Darker colors indicate lower average feature values within a genre, while lighter colors indicate higher values.

Several features exhibited substantial variation across genres. In particular, ‘instrumentalness’, ‘valence’, and ‘acousticness’ displayed some of the largest differences in average values, suggesting strong association with genre identity. This finding supports the idea that genres are not arbitrary labels but are reflected in measurable differences in the acoustic properties of songs. By contrast, features such as ‘loudness’ and ‘mode’ remained relatively consistent across genres, indicating that they may be less useful for distinguishing between genre categories on their own. The presence of both highly variable and relatively stable features suggests that some audio characteristics contribute more strongly than others to genre differentiation.

These results are particularly interesting when compared to the findings from Q1 and Q2. While audio features provided little predictive power for song popularity, they appear to exhibit much clearer structure when examined by genre. This suggests that audio features may be more useful for identifying and classifying musical styles than for predicting commercial success. Consequently, features such as ‘instrumentalness’, ‘valence’, and ‘acousticness’ may warrant further investigation as potential predictors in future genre-classification analyses.

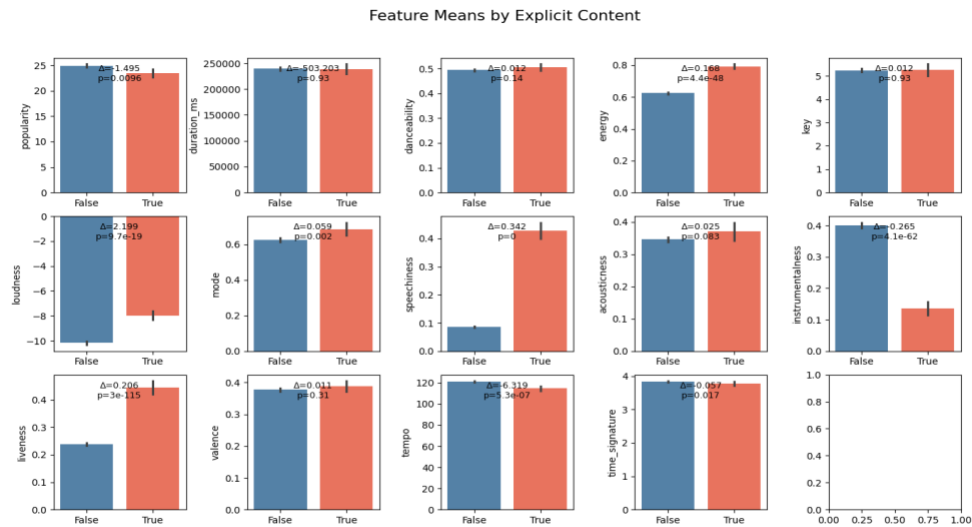


Figure 5

Q4: While several features, notably duration_ms, danceability, key, acousticness, and valence, did not show statistically significant differences between the two groups, separating songs between explicit and non-explicit status did reveal measurable differences in a number of audio features, as shown in Figure 5.

The strongest effect of the filtering found was in the graph for instrumentalness (middle-right), which averaged significantly lower within the explicit songs pool suggesting that explicit tracks tend to be more vocal-driven; that hypothesis is also driven by observing speechiness (middle-right graph) average scores simultaneously and substantially increased for explicit music (net change of 34.2 percentage points).

To extend the analysis beyond audio characteristics, non-audio features were also examined to assess whether explicit labeling also corresponded to broader differences between groups. The expanded analysis revealed a statistically significant difference in popularity between explicit and non-explicit songs ($p = 0.0096$ passes 95% confidence interval check for the results), the first statistically significant relationship between popularity and any predictor observed across the full analysis, including earlier feature-based models in Figures 1-3, but further testing is required to fully examine the extent of this correlation and its potential predicting power.

Overall, explicit labeling appears to be associated with differences in both audio features and non-audio outcomes, suggesting that it captures variation not fully explained by acoustic properties alone. The results reveal a statistically defensible structure not captured by individual audio features alone, and while audio features describe measurable properties of sound, explicit content may reflect additional dimensions that aid in more accurate categorization and feature prediction of music.

Impact and Limitations

The results of this analysis suggest that audio features alone provide limited explanatory power for song popularity. While these features were useful for identifying differences between genres and between explicit and non-explicit music, they were largely unsuccessful at predicting popularity, even when multiple (and nonlinear) machine learning models and additional metadata were incorporated. In spite of that result, the apparent relationship between explicit content and popularity found from the results of Q4 highlights the potential role of categorical/contextual variables in shaping musical success as the first (and only) variable in my analysis to share some statistically significant correlation with the popularity score. It is this finding that sparks my hypothesis for future research into how popularity may be influenced not only by measurable acoustic properties, but broader cultural, social, and contextual factors that are not directly captured in the dataset.

With that context, it should be noted that several limitations should be considered when interpreting these results. The original combined dataset contained over 84,000 songs, but substantial cleaning and filtering were required to obtain a reliable analysis-ready sample. The final dataset was restricted to the 20 most common genres and 500 most common artists, which improved data quality and reduced sparsity but may limit the generalizability of the findings to less common genres, emerging artists, and niche communities. The results of my analysis may be useful to researchers, music analysts, and recommendation-system developers interested in understanding the relationship between musical characteristics and listener behavior; it demonstrates that audio features may not be effective for describing commercial success but perhaps could be applied to describing and classifying music.

Another significant limitation concerns the nature of the data itself. Many of the numerical variables used throughout this analysis were already heavily processed before becoming publicly available. While variables such as ‘duration’ and ‘tempo’ have relatively straightforward definitions, others such as ‘danceability’, ‘energy’, ‘valence’, ‘acousticness’, and ‘instrumentalness’ are algorithmically derived metrics generated by Spotify and/or subsequent data processors. The exact methods used to calculate these values are not fully disclosed, making it difficult to evaluate how accurately they represent the musical concepts they are intended to measure. As a result, these variables should be understood as operational interpretations of musical attributes rather than direct measurements of those attributes.

This introduces an important source of uncertainty into the analysis. Rather than working with raw musical information, the models are built from features that have already been filtered, transformed, and interpreted by Spotify’s systems and then incorporated into publicly available datasets by third-party publishers. Consequently, the results describe relationships within this derived feature space rather than definitive relationships within music itself. Any conclusions drawn from the analysis should therefore be interpreted within the context of the available data and its underlying assumptions.

The analysis is also limited by Spotify's publicly available data. The most informative data for understanding popularity would likely include user behavior, recommendation-system outputs, listening patterns, playlist placement, engagement metrics, and other signals used internally by the company. However, much of this information is proprietary, containing sensitive user data, and making such data publicly available would raise significant privacy concerns for the users of the platform and could expose details of Spotify's recommendation algorithms. Therefore, while access to these variables would likely improve predictive performance, restricting access is both understandable and appropriate.

For these reasons, I highly stress that the conclusions of this study be interpreted as, at best, evidence of patterns within publicly available Spotify data, rather than definitive explanations of why songs become popular (or a lack thereof). The findings of my project identify associations and trends, but they should not be used to make strong causal claims or high-confidence predictions about future commercial success of an artist, music, or any related entity.

Challenges

- **Multiple Datasets:** Successfully merged three Spotify datasets using a composite key (artist, track name) and performed substantial cleaning, deduplication, filtering, and feature engineering before conducting exploratory analysis.
- **Machine Learning:** Implemented and evaluated multiple machine learning models, including Lasso Regression, Random Forest Regression, and Multilayer Perceptron (MLP) Regression, to investigate the predictive power of audio features, genre, and artist information on song popularity.
- **New Library:** Utilized both scikit-learn and statsmodels throughout the project. Scikit-learn was used for machine learning model development, preprocessing, and performance evaluation, while statsmodels was used to conduct Ordinary Least Squares (OLS) regression analyses and statistical significance testing.

Work Plan Evaluation

- **Setup and Data Organization (Finished May 4)**
 - o Completed successfully. Datasets were collected, evaluated, and finalized prior to submission of Part 1.
- **Data Preparation and Cleaning Phase (Finished May 8)**
 - o Completed successfully. Cleaning and merging the datasets led to several adjustments in how variables were represented, particularly with respect to genre categorization and duplicate handling, but the phase remained largely on schedule.
 - o Added a genre-standardized popularity score (popularity_z) on 5/23 to improve comparability across genres and support later visualization and modeling tasks.

- **Exploratory Analysis Phase Relative to Popularity (Finished May 20)**
 - Completed successfully, with exploratory analysis code organized within the project repository.
 - Updated artist filtering criteria on 5/13 to retain only the top 500 most frequently occurring artists, improving data consistency while preserving meaningful variation.
 - Re-ran analyses and updated results on 5/23 after expanding the genre subset from the top 5 genres to the top 20 genres, providing a more representative view of the dataset.
- **Genre and Explicit Content Analysis Phase (Finished May 30)**
 - Completed successfully. Notes on adjustments during this phase made in prior phase' notes with dates.
- **Writing and Synthesis Phase (Finished June 5)**
 - Revised interpretations throughout the project as additional analyses were completed, ensuring that conclusions reflected the final cleaned dataset and updated modeling results.
 - Returned to the Q1 analysis and implemented statistical significance testing to strengthen conclusions regarding the relationship between audio features and popularity.
 - Final completion occurred one day later than originally estimated. Following the project presentation, visualizations were revised to improve clarity and interpretation, as the final written report was expected to receive more detailed scrutiny than the presentation materials.

Testing

The code for this research project was tested through a combination of iterative manual inspection, visualization-based validation (graphs), and structured assertion checks throughout the preprocessing, analysis, and modeling pipeline. During data cleaning and merging, I repeatedly examined the dataset structure by tracking dimensions, column names, data types, missing-value counts, and sample observations. These checks ensured that merge operations, renaming steps, type conversions, filtering procedures, and aggregation logic behaved as expected and preserved key variables throughout transformation stages between running RQ files.

Visualization also served as a key validation tool. Scatterplots, distribution plots, and feature comparisons were used to assess whether relationships between audio features and popularity were reasonable and whether preprocessing decisions produced coherent and interpretable patterns, as well as supported decisions related to genre grouping, filtering, and feature standardization, and helped identify noise in global trends.

In an effort to make sure my results were in a controlled environment and reproducible, I created copies of the dataset prior to major transformations of the data (and such transformations were performed on said copies to leave the original cleaned dataset untouched). This allowed preprocessing and feature engineering steps to be tested without altering the original cleaned dataset and ensured that downstream analyses remained consistent across runs. The project was also organized into separate modules corresponding to each research question, all of which imported shared preprocessing components from a central setup file and the testing file. This modular structure enabled independent execution of each stage while maintaining a consistent data foundation.

In addition to manual inspection and visualization, I utilized assertion-based tests as a systematic mechanism for tracking the state of the dataset throughout each research question. These checks, often compounded together with function calls within one another for compartmentalization, were designed to ensure reproducibility and confirm that the data contained all intended variables at each stage of processing. Assertions were used to validate dataset integrity (shape consistency, missing-value absence, etc.), confirm correct feature selection, and verify that transformations such as filtering, scaling, and genre restriction behaved as expected. Instead of just serving as the primary testing strategy, these assertions functioned as checkpoints that provided immediate feedback on whether the dataset remained in a valid and expected state during execution.

Citations and Collaboration

This project was completed independently without collaboration from other students.

Datasets

- Spotify Analysis and Visualization Dataset (Kaggle): <https://www.kaggle.com/datasets/sarametawea/spotify-analysis-and-visualization>
- Spotify Tracks Dataset (Kaggle): <https://www.kaggle.com/datasets/maharshipandya/-spotify-tracks-dataset>
- Spotify Tracks Attributes and Popularity Dataset (Kaggle): <https://www.kaggle.com/datasets/melissamonfared/spotify-tracks-attributes-and-popularity>

Libraries Utilized:

- Pandas
- NumPy
- Matplotlib
- Seaborn
- Statsmodels
- Scikit-learn

All data cleaning, analysis, visualization, model development, interpretation, and conclusions were completed independently.